

Manufacturing Documentation
Viscous Damping Device for Parkinson's Tremor
Senior Design I – Spring 2025

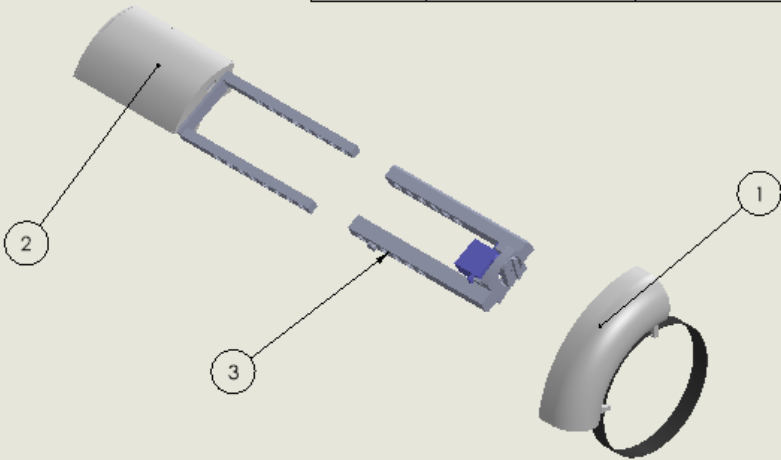


Bill of Materials:

Part Number	Description	Quantity
1	Fluid Dampening Brace	1
2	Detection System	1
3	Oil System	1
4	Magnet Servo system	1
5	Fluid Housing and Rail	1
6	Baffle	1
7	Fluid Cap	2
8	Magnet Holder	1
9	Servo Beam	1
10	Wrist Beam	1
11	Telescoping Pin	1
12	E-bay Case Assembly	1
13	Ebay Lid	1
14	Wrist Strap	1
15	Magnet	1
16	Forearm Strap	2
17	Servo	1
18	MPU6050	1
19	AMS1117-3.3	1
20	ESP32-MINI-1	1
21	TP4056	1
22	MT3608	1
23	3.7V Li-Io Battery	1
24	Wires	15
25	Canola Oil	1

Fluid Dampening Brace

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Part 3	Fluid housing system	1
2	Part 2	Detection system	1
3	Part 4	Magnet Servo system	1



DESIGNER: ME 574 Team 3

DEFAULT TOLERANCE (IN)

.X: ± 0.1

.XX: ± 0.05

.XXX: ± 0.001

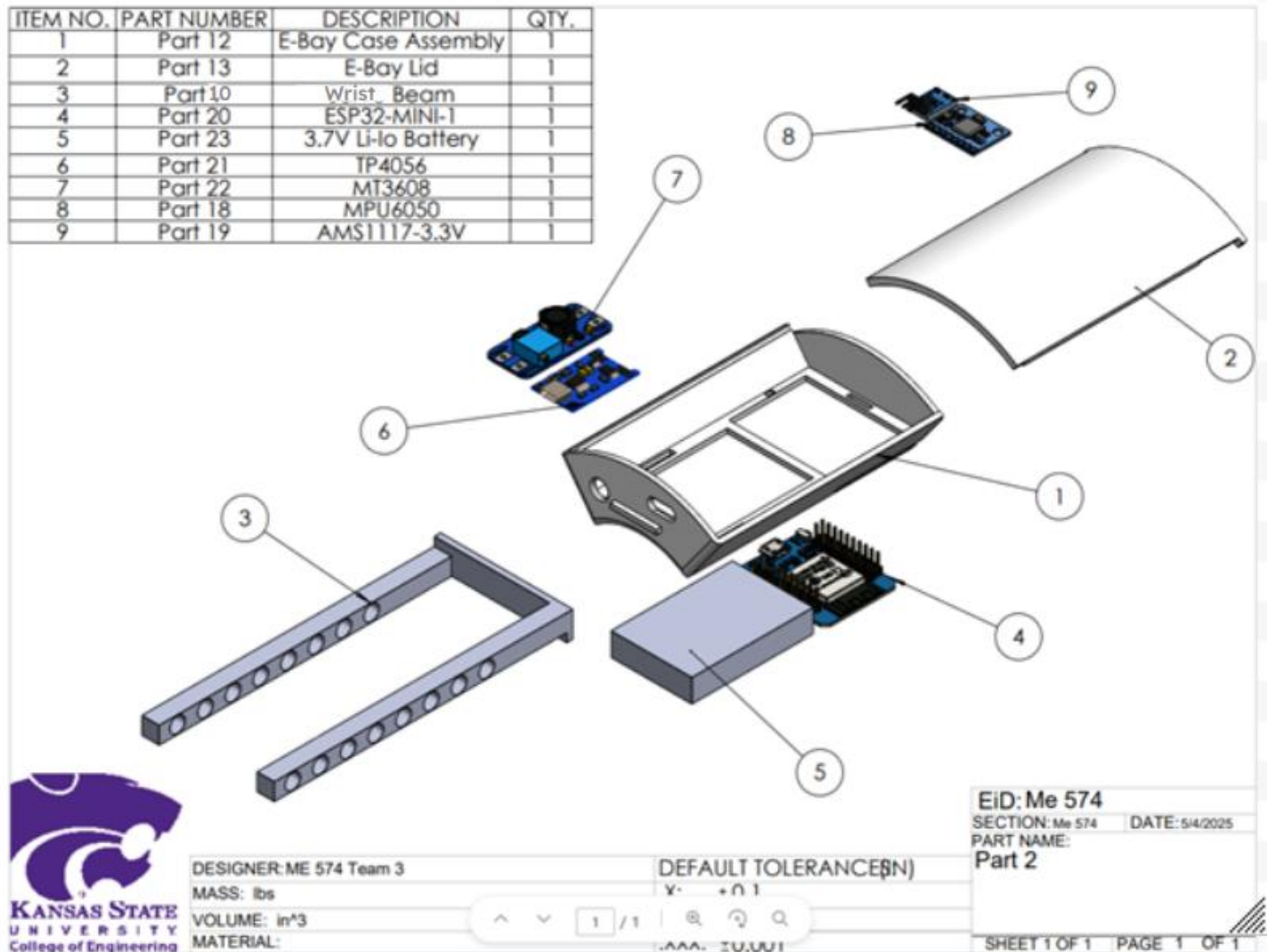
EiD: Me 574

SECTION: Me 574 DATE: 5/8/2025

PART NAME:

Part 1

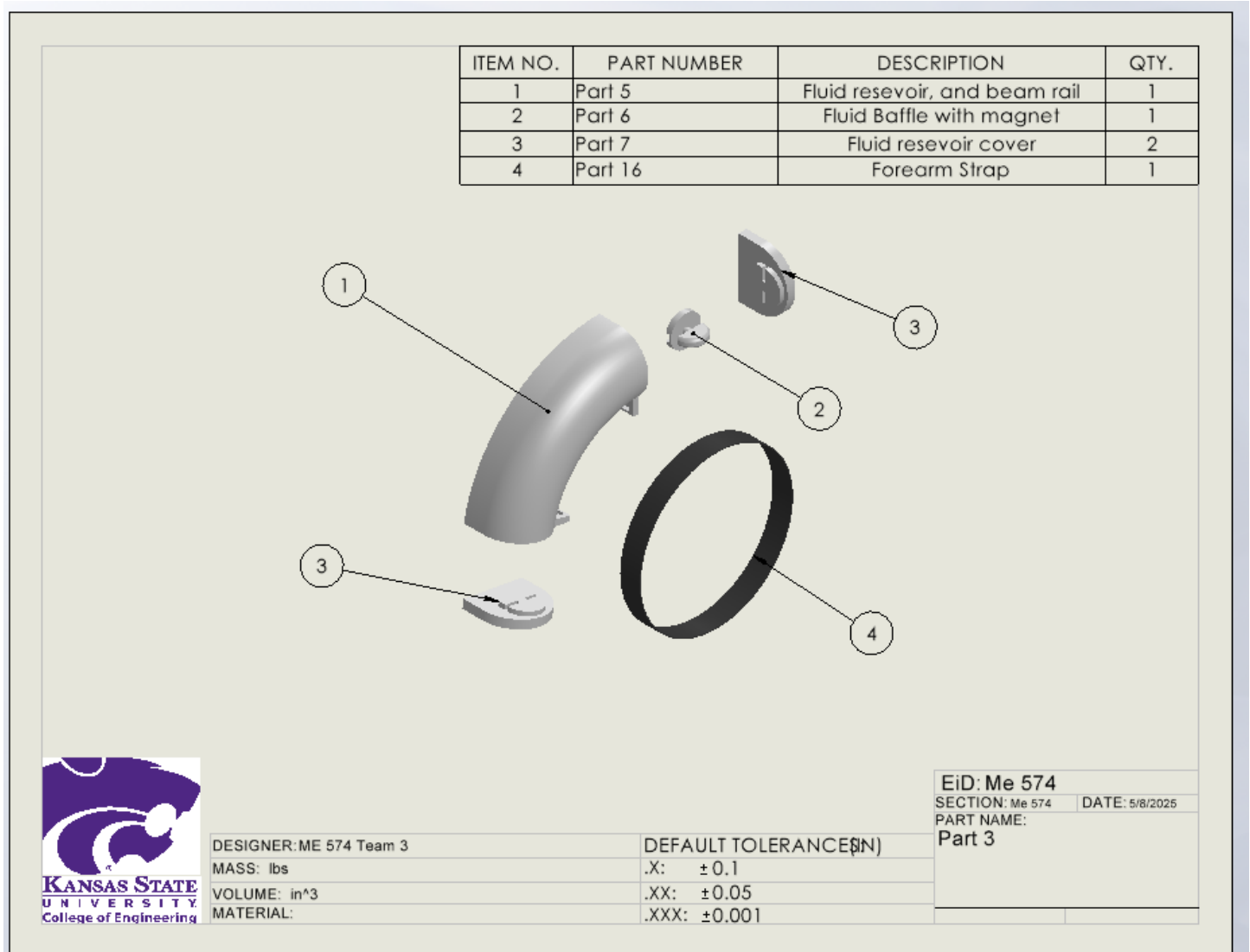
Detection System



Additional Assembly instructions

- Servo wires will be fed through parts 10 and 12
- Adhesive should be applied to the connection between parts 10 and 12

Fluid Housing System

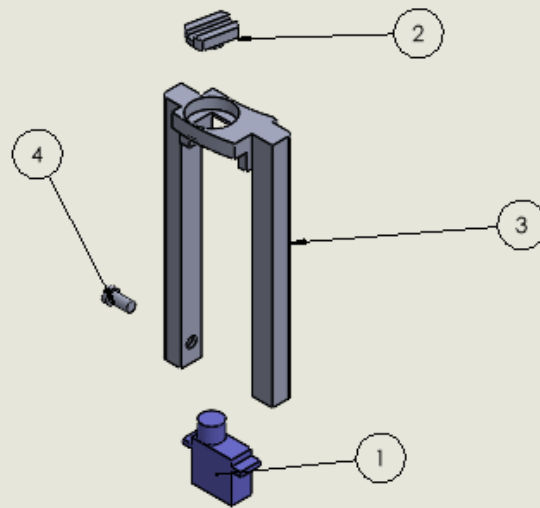


Additional Assembly instructions

- Part 16 (wrist strap) will be fed through loops in part 5.
- Seal one of the fluid caps (part 7)
- Orientate baffle into the reservoir so that the magnet is tangent with reservoir arc.
- Pour in canola oil then seal reservoir with part 7

Magnet Servo System

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Part 17	Servo	1
2	Part 8	Magnet Holder	1
3	Part 9	Servo Beam	1
4	Part 11	Teloscopying pin	1



DESIGNER: ME 574 Team 3

MASS: lbs

VOLUME: in³

MATERIAL:

DEFAULT TOLERANCE (IN)

.X: ± 0.1

.XX: ± 0.05

.XXX: ± 0.001

EiD: Me 574

SECTION: Me 574 DATE: 5/8/2025

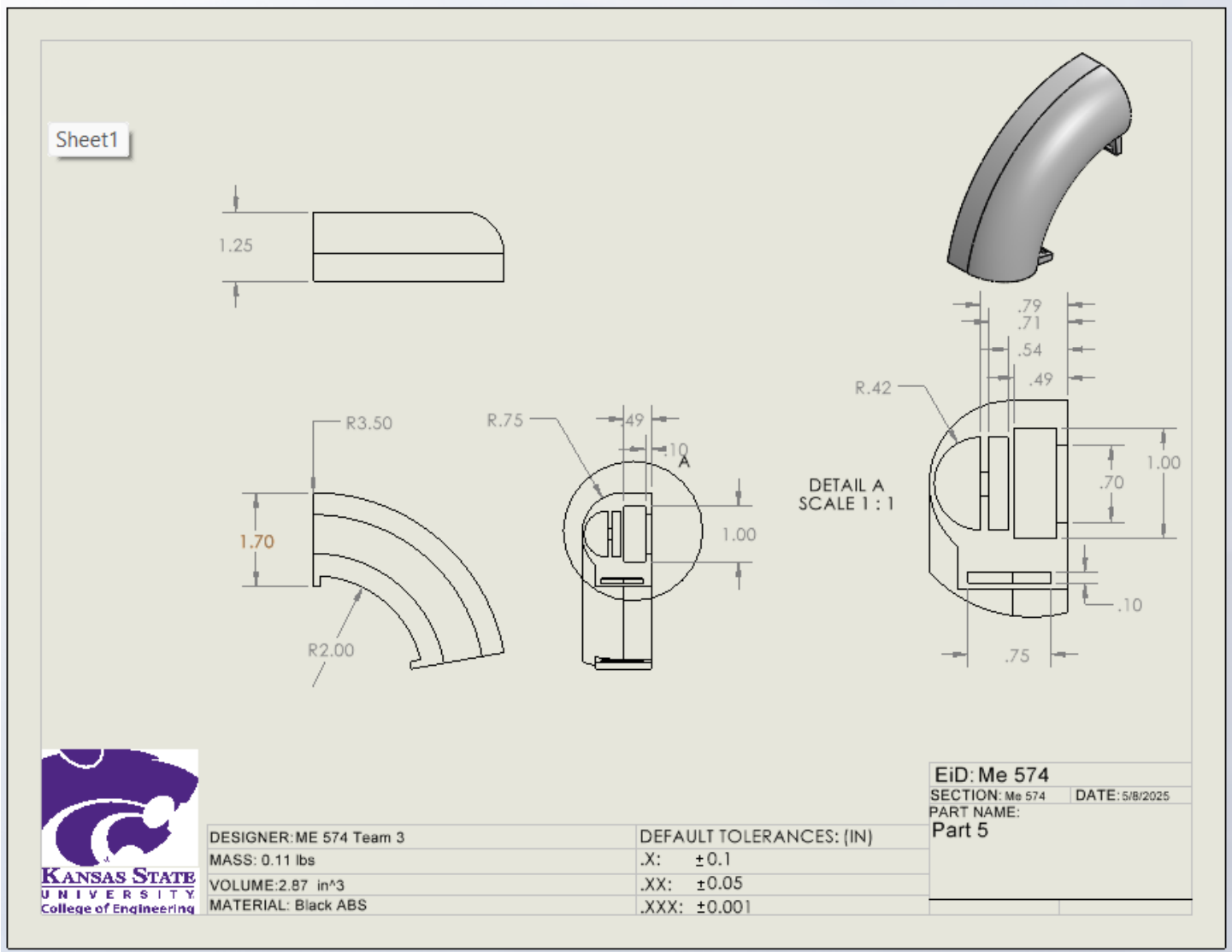
PART NAME:

Part 4

Additional assembly instructions

- Feed servo wire into hole in part 9

Fluid Housing



Injection Molding 1 Part

ITAR/Export Controlled: No

Want help? [Schedule a free design review](#)

Part 5.SLDPRT

1375-3116-001

CAD Revision: 1

Mold Life: Unlimited (On-demand Manufacturing)

1 Cavity

ABS : Cyclic MG47 (Black)

Black (Original Material Color)

Cosmetic: PM-F0

Non-Cosmetic: PM-F0

X: 4.499in Y: 2.121in Z: 1.250in

Machining Tolerance: +/- 0.003 in. (0.076 mm)

Material Tolerance: +/- 0.002 in./in. (0.002 mm/mm)

[See inspections options](#)

⚠ This part needs your attention: ⓘ

Sample Quantity

- 25 +

[View Analysis](#)

[Configure Part](#)

[Upload Revision](#)

[Part Options](#)

25 Parts @ \$4.21	\$105.25
Capability Study	\$0.00
CTQ Partial FAI	\$0.00
Mold	\$12,005.00
Total	\$12,110.25

Order by:
Fri 4:00 PM

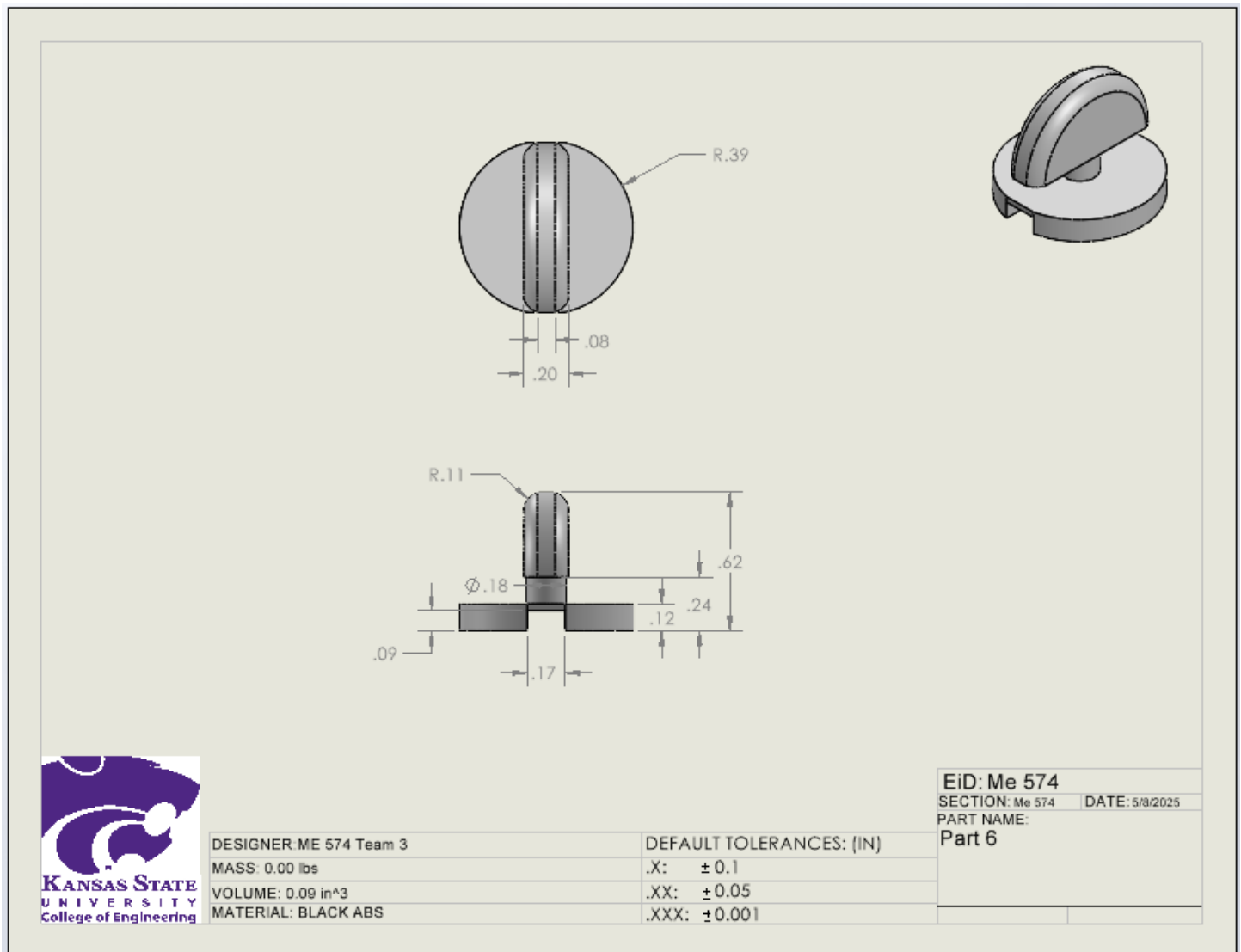
Receive by:
Tue, Jun 3

Expedite Wed, May 21	Expedite Tue, May 27	Expedite Thu, May 29	Standard Tue, Jun 3
+ \$5,042.00	+ \$3,001.00	+ \$2,041.00	

Tooling Cost										
Move?	Item	Snap Shot	Part Number/Name	Material?	Cavities?	Color?	Surface Finish?	Sample Quantity	Tooling Lead Time?	Tooling Price with samples
	1		Part 5.slp	ABS Chimei PA-757 Natural Advanced Material Search	1 ▾	NATURAL COLOR ▾	Standard No Machine Mar ▾	5	15 Business Days	\$5,605

Part Cost							
Secondary Process?		Item	Part Number, Size & Weight	Part Quantity	Part Price	Estimated Price Ea. w/ Air Shipping?	Estimated Price Ea. w/ Ocean Shipping?
Click to Select ▾		1	Part 5.slp 81.0x31.7x87.5mm 0.0494kg	2500  Updating...	\$0.77	\$1.63	\$1.72

Baffle



Additional Assembly instructions

- Place Neodymium Bar Magnets 60 x 10 x 3 mm into slot in baffle

Injection Molding1 Part

ITAR/Export Controlled: No

Part 6.SLDPRT

1807-0036-001

CAD Revision: 1

Mold Life: Unlimited (On-demand Manufacturing)

1 Cavity

ABS : Cycolac MG47 (Black)

Black (Original Material Color)

Cosmetic: PM-F0

Non-Cosmetic: PM-F0

X: 0.760in Y: 0.620in Z: 0.778in

Machining Tolerance: +/- 0.003 in. (0.076 mm)

Material Tolerance: +/- 0.002 in./in. (0.002 mm/mm)

See inspections options

This part needs your attention:

View Analysis

Configure Part

Upload Revision

Part Options

Sample Quantity

-

25

+

See Volume Pricing

25 Parts @ \$2.69

\$67.25

Capability Study

\$0.00

CTQ Partial FAI

\$0.00

Mold

\$6,290.00

Total

\$6,357.25

Order by:

Fri 4:00 PM

Receive by:

Tue, Jun 3

Expedite

Sat, May 17

+ \$3,145.00

Expedite

Mon, May 19

+ \$3,145.00

Expedite

Wed, May 21

+ \$2,642.00

Expedite

Tue, May 27

+ \$1,572.00

Expedite

Thu, May 29

+ \$1,069.00

Standard

Tue, Jun 3

Tooling Cost										
Move?	Item	Snap Shot	Part Number/Name	Material?	Cavities?	Color?	Surface Finish?	Sample Quantity	Tooling Lead Time?	Tooling Price with samples
	1		Part 6 NOT FINISHED.stp	ABS Chimei PA-757 Natural Advanced Material Search	1	NATURAL COLOR	Standard No Machine Mar	5	15 Business Days	\$2,934

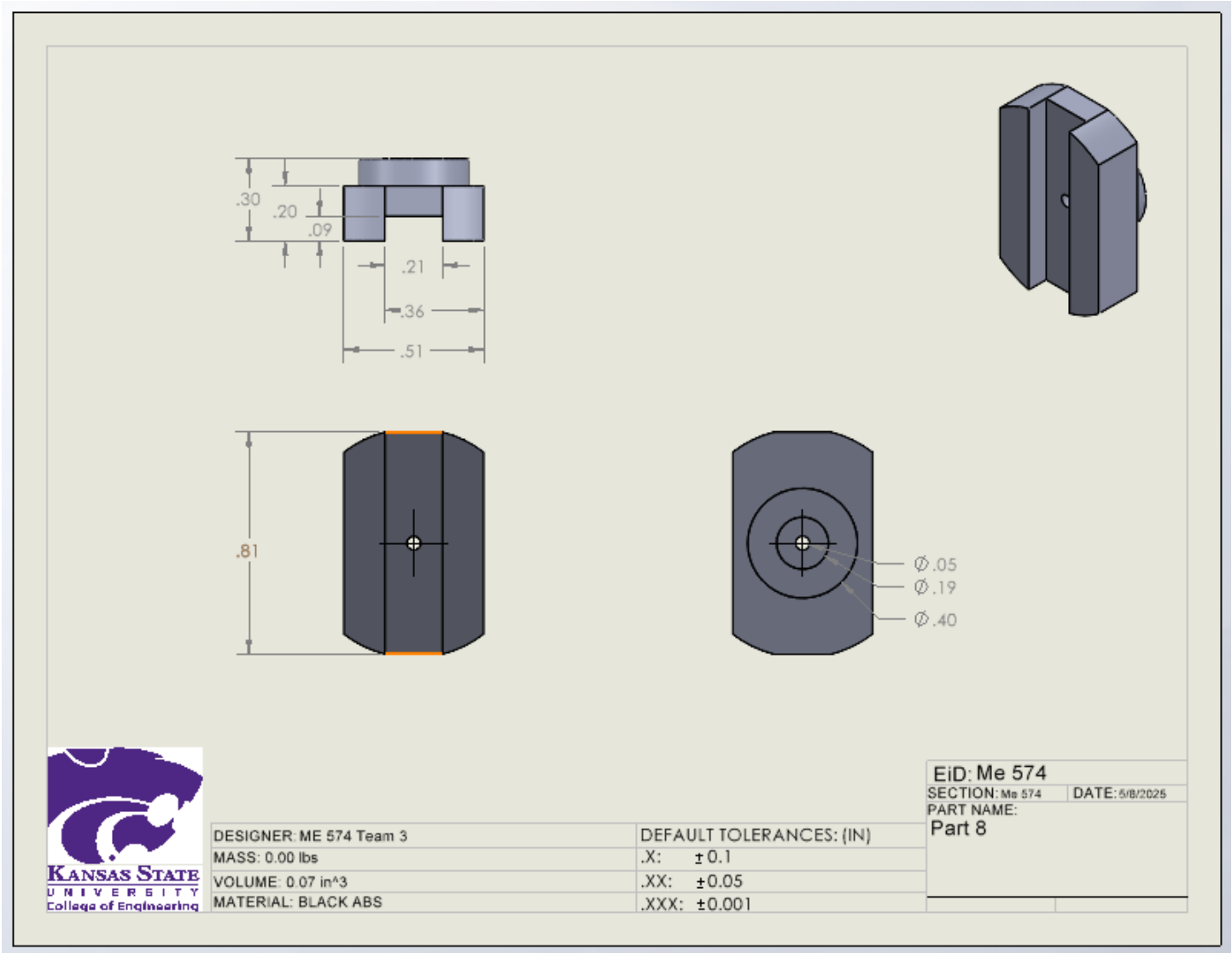
Part Cost							
Secondary Process?		Item	Part Number, Size & Weight	Part Quantity	Part Price	Estimated Price Ea. w/ Air Shipping?	Estimated Price Ea. w/ Ocean Shipping?
Click to Select		1	Part 6 NOT FINISHED.stp 19.8×15.7×19.3mm 0.0016kg	2500 GO Updating...	\$0.28	\$0.40	\$1.13

Tooling Cost										
Move	Item	Snap Shot	Part Number/Name	Material	Cavities	Color	Surface Finish	Sample Quantity	Tooling Lead Time	Tooling Price with samples
	1		Part 7.stp	ABS Chimel PA-757 Natural Advanced Material Search	1	NATURAL COLOR	Standard No Machine Mar	5	15 Business Days	\$4,762

Part Cost

Secondary Process	Item	Part Number, Size & Weight	Part Quantity	Part Price	Estimated Price Ea. w/ Air Shipping	Estimated Price Ea. w/ Ocean Shipping
Click to Select	1	Part 7.stp 31.8x7.6x38.1mm 0.0059kg	2500 GO	\$0.37	\$0.57	\$1.24

Magnet Holder



Additional Assembly instructions

- Place Neodymium Bar Magnets 60 x 10 x 3 mm into slot in part 8

Injection Molding 1 Part

ITAR/Export Controlled: No



Part 8.SLDPRT

1948-0559-001

CAD Revision: 1

Mold Life: Unlimited (On-demand Manufacturing)

1 Cavity

ABS : Cycolac MG47 (Black)

Black (Original Material Color)

Cosmetic: PM-F0

Non-Cosmetic: PM-F0

X: 0.810in Y: 0.510in Z: 0.300in

Machining Tolerance: +/- 0.003 in. (0.076 mm)

Material Tolerance: +/- 0.002 in./in. (0.002 mm/mm)

See inspections options

⚠

This part needs your attention: ⓘ

View Analysis

Configure Part

Upload Revision

Part Options ▾

Sample Quantity

-

25

+

[See Volume Pricing](#)

25 Parts @ \$2.23

\$55.75

Capability Study

\$0.00

CTQ Partial FAI

\$0.00

Mold

\$2,320.00

Total

\$2,375.75

Order by:

Today 4:00 PM

Receive by:

Fri, May 16

Expedite

Thu, May 8

+ \$4,640.00

Expedite

Mon, May 12

+ \$2,320.00

Expedite

Wed, May 14

+ \$1,160.00

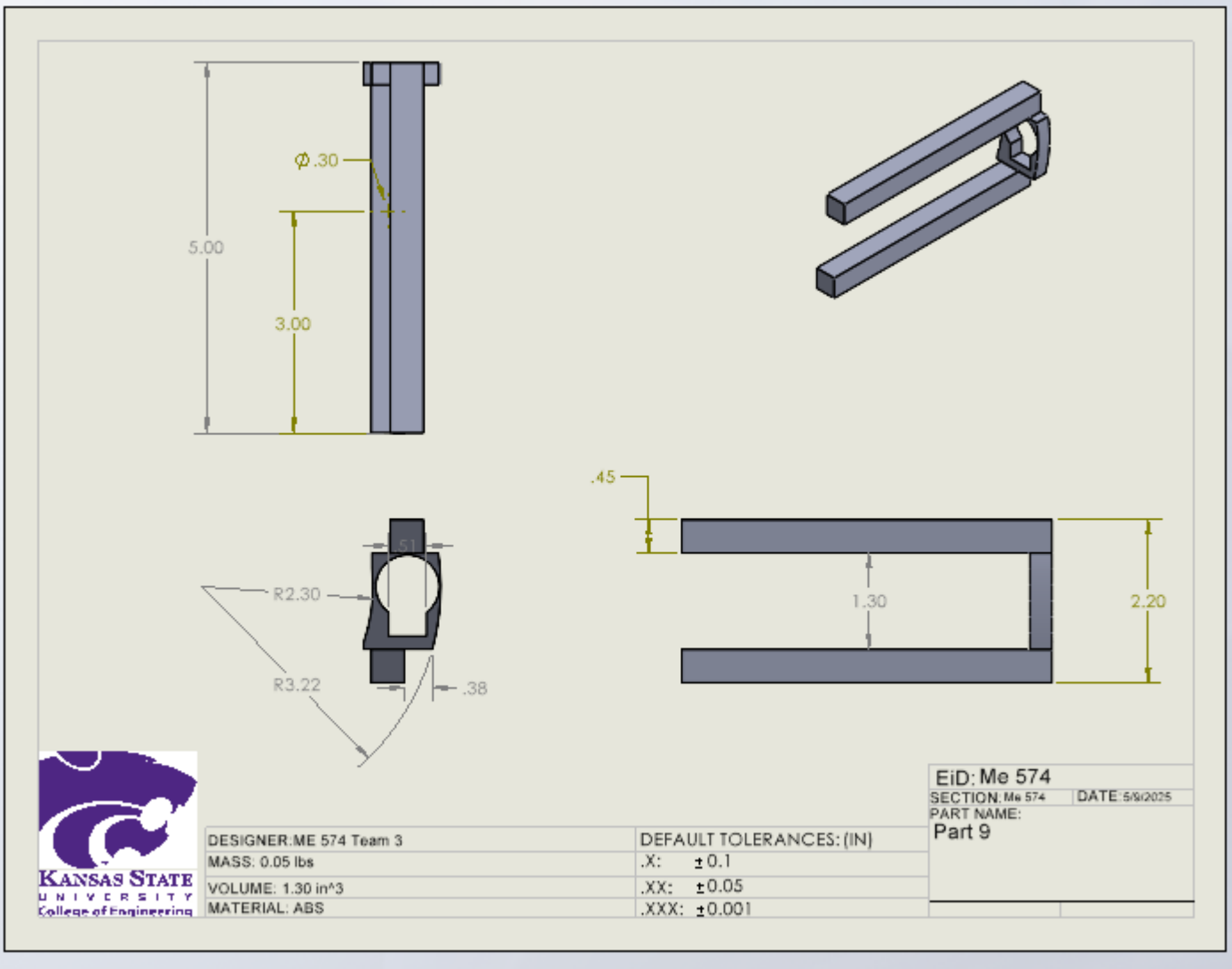
Standard

Fri, May 16

Tooling Cost										
Move?	Item	Snap Shot	Part Number/Name	Material?	Cavities?	Color?	Surface Finish?	Sample Quantity	Tooling Lead Time?	Tooling Price with samples
	1		Part 8.stp	ABS ChimeI PA-757,Natural Advanced Material Search	1 ▾	NATURAL COLOR ▾	Standard No Machine Mar ▾	5	15 Business Days	\$2,934

Part Cost							
Secondary Process?		Item	Part Number, Size & Weight	Part Quantity	Part Price	Estimated Price Ea. w/ Air Shipping?	Estimated Price Ea. w/ Ocean Shipping?
Click to Select ▾		1	Part 8.stp 13.0×7.6×20.6mm 0.0013kg	2500 GO	\$0.28	\$0.39	\$1.13

Servo Beam



Injection Molding 1 Part

ITAR/Export Controlled: No

Part 9.SLDPRT

1654-9562-001

CAD Revision: 1

Mold Life: Unlimited (On-demand Manufacturing)

1 Cavity

ABS : Cyclic MG47 (Black)

Black (Original Material Color)

Cosmetic: PM-F0

Non-Cosmetic: PM-F0

X: 5.000in Y: 2.200in Z: 1.042in

Machining Tolerance: +/- 0.003 in. (0.076 mm)

Material Tolerance: +/- 0.002 in./in. (0.002 mm/mm)

See inspections options

This part needs your attention: ⓘ

View Analysis

Configure Part

Upload Revision

Part Options ▾

Sample Quantity

- 25 +

See Volume Pricing

25 Parts @ \$3.35

\$83.75

Capability Study

\$0.00

CTQ Partial FAI

\$0.00

Mold

\$9,225.00

Total

\$9,308.75

Order by:

Fri 4:00 PM

Receive by:

Tue, Jun 3

Expedite

Sat, May 17

+ \$4,612.00

Expedite

Mon, May 19

+ \$4,612.00

Expedite

Wed, May 21

+ \$3,874.00

Expedite

Tue, May 27

+ \$2,306.00

Expedite

Thu, May 29

+ \$1,568.00

Standard

Tue, Jun 3

TOOLING COST

Move	Item	Snap Shot	Part Number/Name	Material	Cavities	Color	Surface Finish	Sample Quantity	Tooling Lead Time	Tooling Price with samples
	1		Part 9.slp	ABS Chimei PA-757 Natural Advanced Material Search	1	NATURAL COLOR	Standard No Machine Mar	5	15 Business Days	\$7,460

PART COST

SECONDARY PROCESS	Item	Part Number, Size & Weight	Part Quantity	Part Price	Estimated Price Ea. w/ Air Shipping	Estimated Price Ea. w/ Ocean Shipping
Click to Select	1	Part 9.slp 55.9x26.5x127.0mm 0.0223kg	2500 QD Updating...	\$0.58	\$1.12	\$1.50

Wrist Beam

Technical drawing of the Wrist Beam part. The top view shows a rectangular shape with a width of 2.00 and a height of 1.95. The side view shows a thickness of .29. The end view shows a width of .47 and a height of .02. The bottom view shows a series of holes with a diameter of .20 and a spacing of .08. The overall length of the part is 4.10, with a central section of 4.00. The part is made of ABS material.

3D isometric view of the Wrist Beam part, showing its L-shaped profile and the series of holes along the bottom edge.

DESIGNER: ME 574 Team 3
MASS: 0.02 lbs
VOLUME: 0.55 in³
MATERIAL: ABS

DEFAULT TOLERANCES: (IN)
.X: ±0.1
.XX: ±0.05
.XXX: ±0.001

EiD: Me 574
SECTION: Me 574 **DATE:** 5/9/2025
PART NAME: Part 10

KANSAS STATE UNIVERSITY
College of Engineering

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×



Part 10.SLDPRT
1433-6135-001
CAD Revision: 1
Mold Life: Unlimited (On-demand Manufacturing)
1 Cavity
ABS : Cycolac MG47 (Black)
Black (Original Material Color)
Cosmetic: PM-F0
Non-Cosmetic: PM-F0
X: 2.110in Y: 4.199in Z: 0.640in
Machining Tolerance: +/- 0.003 in. (0.076 mm)
Material Tolerance: +/- 0.002 in./in. (0.002 mm/mm)
[See inspections options](#)
 This part needs your attention: 

Sample Quantity

-

25

+

 [See Volume Pricing](#)

25 Parts @ \$3.96	\$99.00
Capability Study	\$0.00
CTQ Partial FAI	\$0.00
Mold	\$13,495.00
Total	\$13,594.00

Order by:
Fri 4:00 PM
Receive by:
Tue, Jun 3

Expedite	Expedite	Expedite	Standard
Wed, May 21	Tue, May 27	Thu, May 29	Tue, Jun 3
+ \$5,668.00	+ \$3,374.00	+ \$2,294.00	

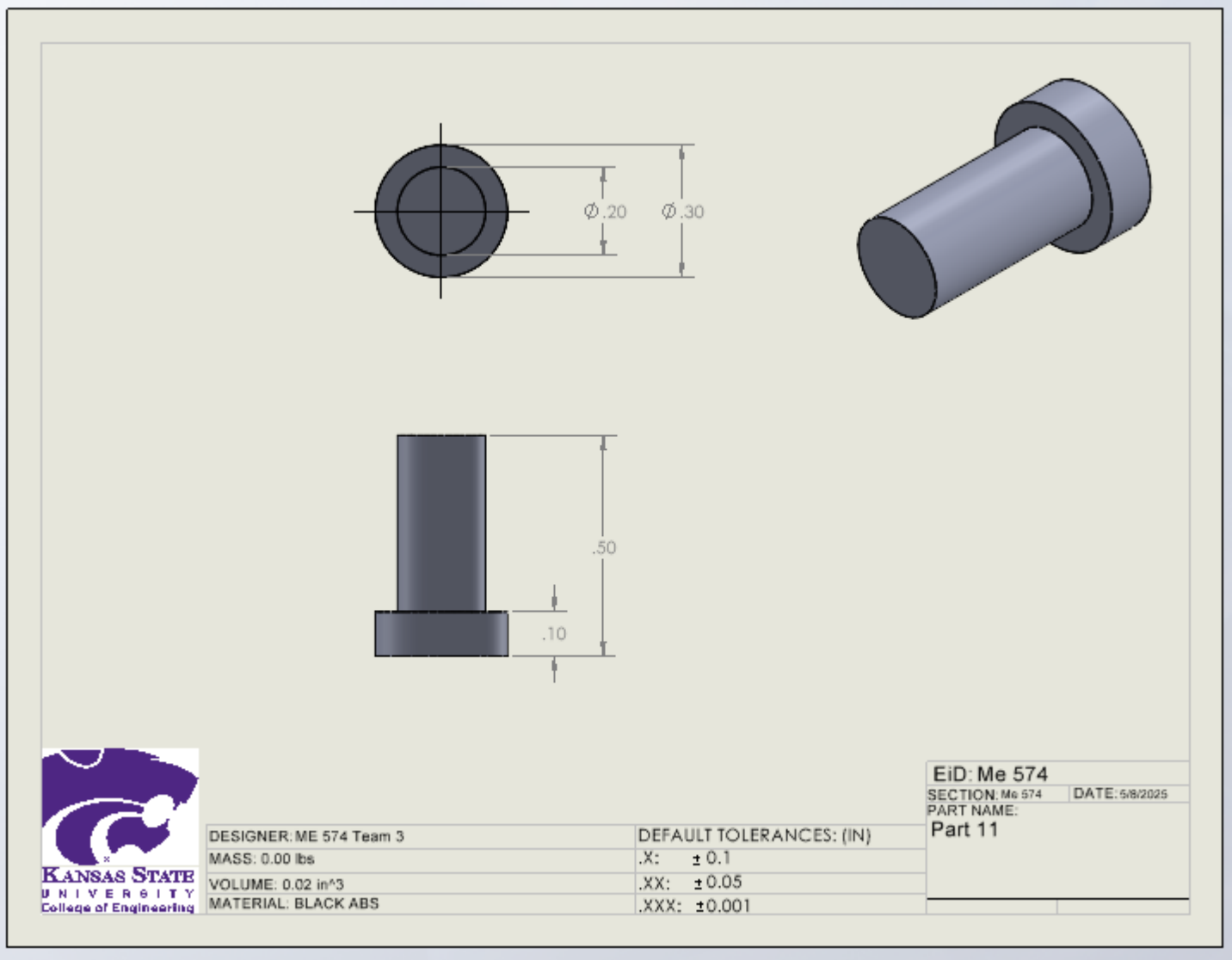
TOOLING COST

Move	Item	Snap Shot	Part Number/Name	Material	Cavities	Color	Surface Finish	Sample Quantity	Tooling Lead Time	Tooling Price with samples	
	1		Part 10.stp	ABS Chimel PA-757 Natural Advanced Material Search	1	NATURAL COLOR	Standard No Machine Ma	5	15 Business Days	\$6,076	

PART COST

SECONDARY PROCESS		Item	Part Number, Size & Weight	Part Quantity	Part Price	Estimated Price Ea. w/ Air Shipping	Estimated Price Ea. w/ Ocean Shipping
Click to Select		1	Part 10.stp 53.6x16.3x196.7mm 0.0095kg	2500 GO Updating...	\$0.42	\$0.72	\$1.31

Telescoping Pin



Injection Molding 1 Part

ITAR/Export Controlled: No



Part 11.SLDPRT

1467-1030-001

CAD Revision: 1

Mold Life: Unlimited (On-demand Manufacturing)

1 Cavity

ABS : Cycolac MG47 (Black)

Black (Original Material Color)

Cosmetic: PM-F0

Non-Cosmetic: PM-F0

X: 0.500in Y: 0.300in Z: 0.300in

Machining Tolerance: +/- 0.003 in. (0.076 mm)

Material Tolerance: +/- 0.002 in./in. (0.002 mm/mm)

[See inspection options](#)

 This part needs your attention: ⓘ

Sample Quantity

- 25 +

[See Volume Pricing](#)

25 Parts @ \$2.21	\$55.25
Capability Study	\$0.00
CTQ Partial FAI	\$0.00
Mold	\$2,315.00
Total	\$2,370.25

Order by:
Today 4:00 PM
Receive by:
Fri, May 16

Expedite
Thu, May 8
+ \$4,630.00

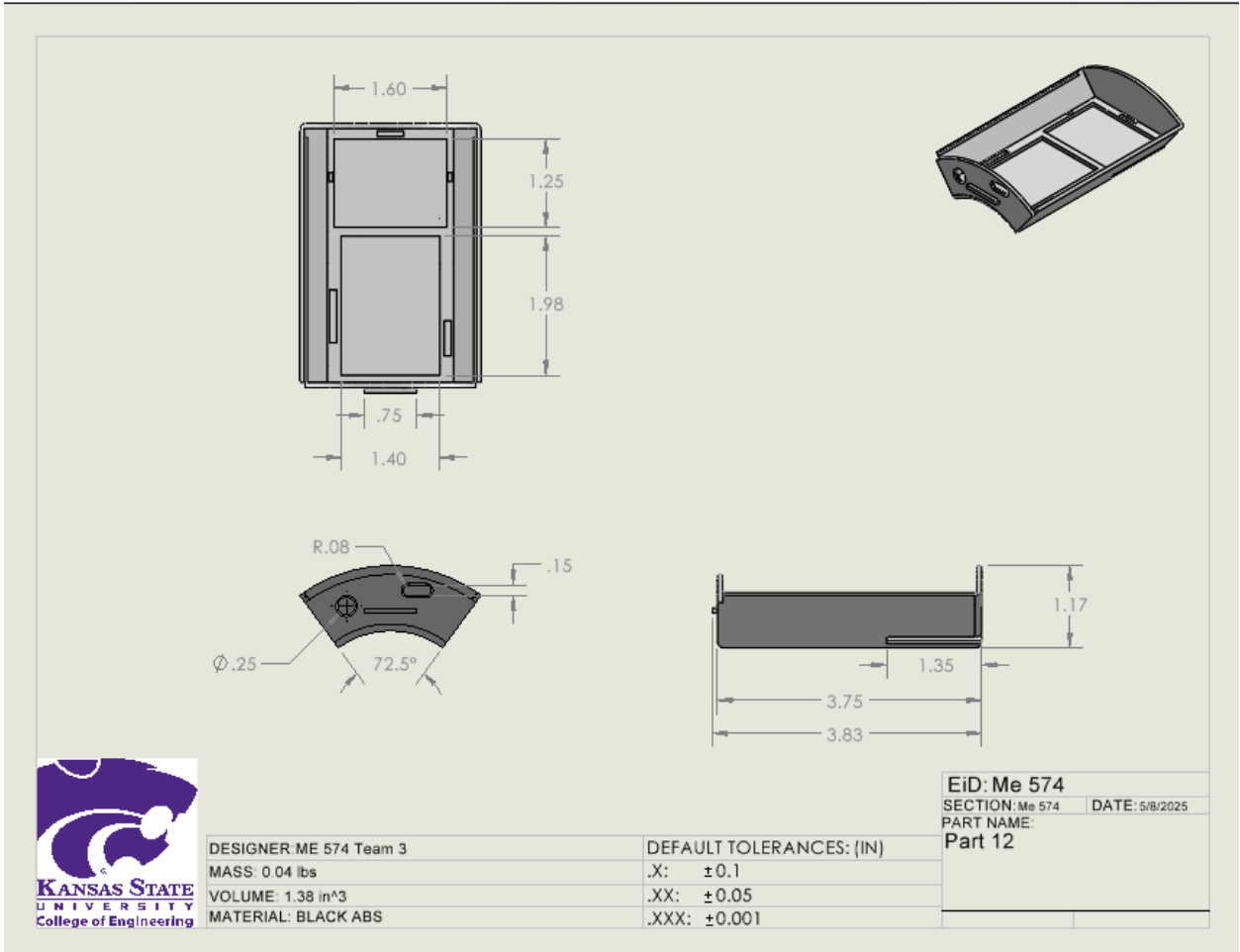
Expedite
Mon, May 12
+ \$2,315.00

Expedite
Wed, May 14
+ \$1,158.00

Standard
Fri, May 16

TOOLING COST										
Move?	Item	Snap Shot	Part Number/Name	Material?	Cavities?	Color?	Surface Finish?	Sample Quantity	Tooling Lead Time?	Tooling Price with samples
	1		Part 11.stp	ABS ChimeI PA-757, Natural Advanced Material Search	1	NATURAL COLOR	Standard No Machine Mar	5	15 Business Days	\$2,934

PART COST							
SECONDARY PROCESS?		Item	Part Number, Size & Weight	Part Quantity	Part Price	Estimated Price Ea. w/ Air Shipping?	Estimated Price Ea. w/ Ocean Shipping?
Click to Select		1	Part 11.stp 12.7×7.6×7.6mm 0.0003kg	2500 GO Updating...	\$0.28	\$0.37	\$1.13



Quote Date: April 9, 2025

Quote 6684-186

Prepared for Kansas State University

Injection Molding 1 Part		ITAR/Export Controlled: No	
	casing.SLDPRT 1179-9154-001 CAD Revision: 1 Mold Life: Limited 1 Cavity ABS : Lustran 248 (White) White (Original Material Color) Cosmetic: PM-F0 Non-Cosmetic: PM-F0 X: 1.978in Y: 0.831in Z: 2.000in Machining Tolerance: +/- 0.003 in. (0.07 ... Material Tolerance: +/- 0.002 in./in. (0.0... See inspections options	Sample Quantity 25	See Volume Pricing
	25 Parts @ \$2.39 \$59.75 Mold \$3,660.00 Total \$3,719.75		

Tooling Cost											
Move	Item	Snap Shot	Part Number/Name	Material	Cavities	Color	Surface Finish	Sample Quantity	Tooling Lead Time	Tooling Price with samples	
	1		casing2.1.stp	ABS Chimel PA-757 Natural Advanced Material Search	1	NATURAL COLOR	Standard No Machine Ma	5	15 Business Days	\$7,285	

Part Cost							
Secondary Process	Item	Part Number, Size & Weight	Part Quantity	Part Price	Estimated Price Ea. w/ Air Shipping	Estimated Price Ea. w/ Ocean Shipping	
Click to Select	1	casing2.1.stp 97.2×65.3×29.6mm 0.0239kg	2500 Go	\$0.59	\$1.11	\$1.50	

Technical drawing of a curved part (Part 13) showing top, side, and isometric views with dimensions and tolerances.

Top View Dimensions:

- Overall width: 2.57
- Inner width: 2.45
- Left side height: 3.68
- Right side height: 3.60

Side View Dimensions:

- Radius: R2.18
- Left side thickness: .13
- Right side thickness: .05

Isometric View: A 3D perspective view of the part showing its curved shape.

Table 1: Design Information

DESIGNER: ME 574 Team 3	DEFAULT TOLERANCES: (IN)
MASS: 0.03 lbs	.X: ± 0.1
VOLUME: 0.95 in ³	.XX: ± 0.05
MATERIAL: BLACK ABS	.XXX: ± 0.001

Table 2: Section and Date Information

EID: Me 574	DATE: 5/8/2025
SECTION: Me 574	
PART NAME:	
Part 13	

Table 3: Sheet and Page Information

SHEET 1 OF 1	PAGE 13 OF 25
--------------	---------------

ITAR/Export Controlled: N

x



25 Parts @ \$3.16	\$79.00
Capability Study	\$0.00
CTQ Partial FAI	\$0.00
Mold	\$13,450.00
Total	\$13,529.00

 This part needs your attention: 

Tooling Cost											
Move?	Item	Snap Shot	Part Number/Name	Material?	Cavities?	Color?	Surface Finish?	Sample Quantity	Tooling Lead Time	Tooling Price with samples	
	1		Lid stp	ABS Chimel PA-757 Natural Advanced Material Search	1	NATURAL COLOR	Standard No Machine Mar	5	15 Business Days	\$7,409	

Forearm Strap



Note: 36x2x0.1 in utility strap with clip or equivalent



DESIGNER: ME 574 Team 3

MASS: lbs

VOLUME: in^3

MATERIAL:

DEFAULT TOLERANCE (in)

.X: ±0.1

.XX: ±0.05

.XXX: ±0.001

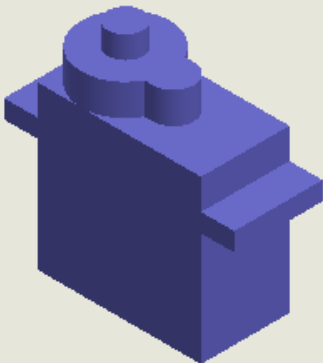
EiD: Me 574

SECTION: Me 574 DATE: 5/9/2025

PART NAME:

Part 16

Servo



Note: 9g servo TIANKONGRC brand or equivalent



DESIGNER:ME 574 Team 3

MASS: lbs

VOLUME: in^3

MATERIAL:

DEFAULT TOLERANCES (in)

.X: ±0.1

.XX: ±0.05

.XXX: ±0.001

EiD: Me 574

SECTION:Me 574

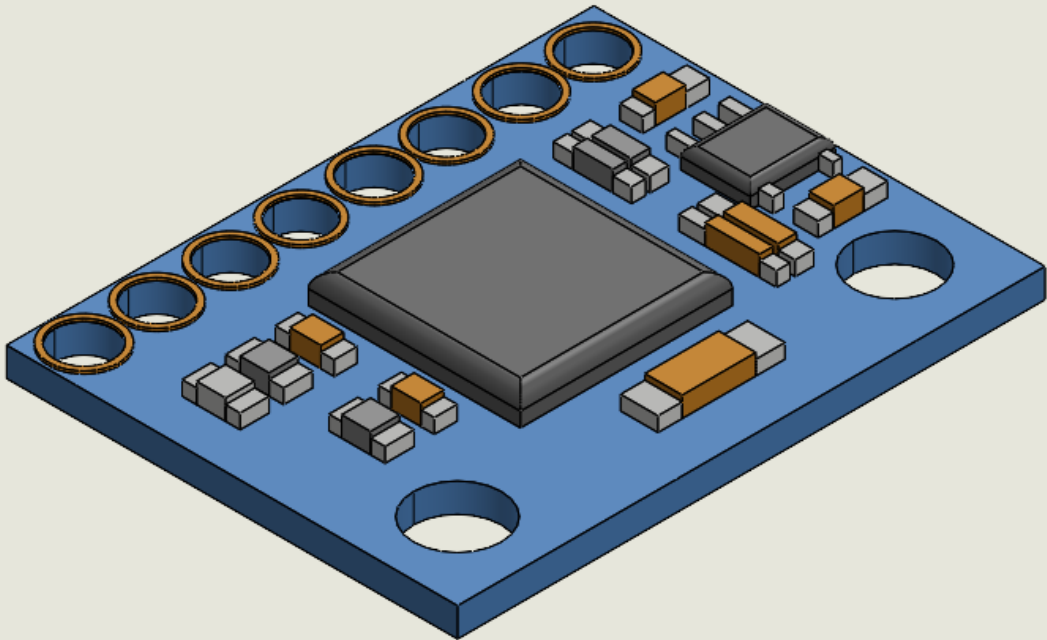
DATE: 5/9/2025

PART NAME:

Part 17

MPU6050

MPU6050 ACCELEROMETER
*OR EQUIVALENT



KANSAS STATE
UNIVERSITY
College of Engineering

DESIGNER: ME 574 Team 3

MASS: lbs

VOLUME: in^3

MATERIAL:

DEFAULT TOLERANCE (IN)

.X: ±0.1

.XX: ±0.05

.XXX: ±0.001

EiD: Me 574

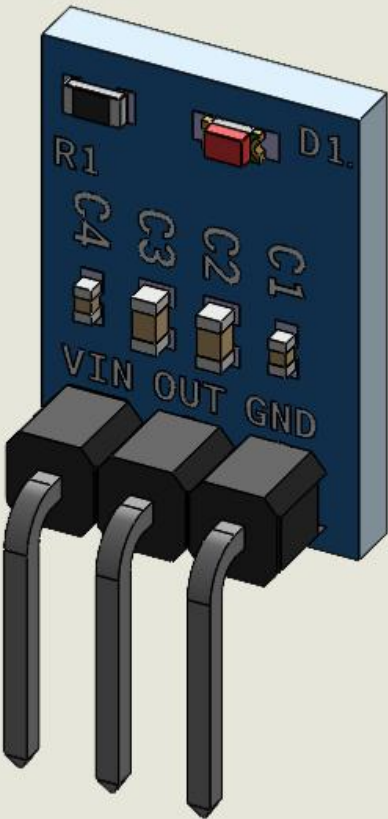
SECTION: Me 574

DATE: 5/4/2025

PART NAME:

Part 18

AMS1117-3.3



AMS1117-3.3V
*OR EQUIVALENT



KANSAS STATE
UNIVERSITY
College of Engineering

DESIGNER: ME 574 Team 3

MASS: lbs

VOLUME: in^3

MATERIAL:

DEFAULT TOLERANCE(\$N)

.X: ±0.1

.XX: ±0.05

.XXX: ±0.001

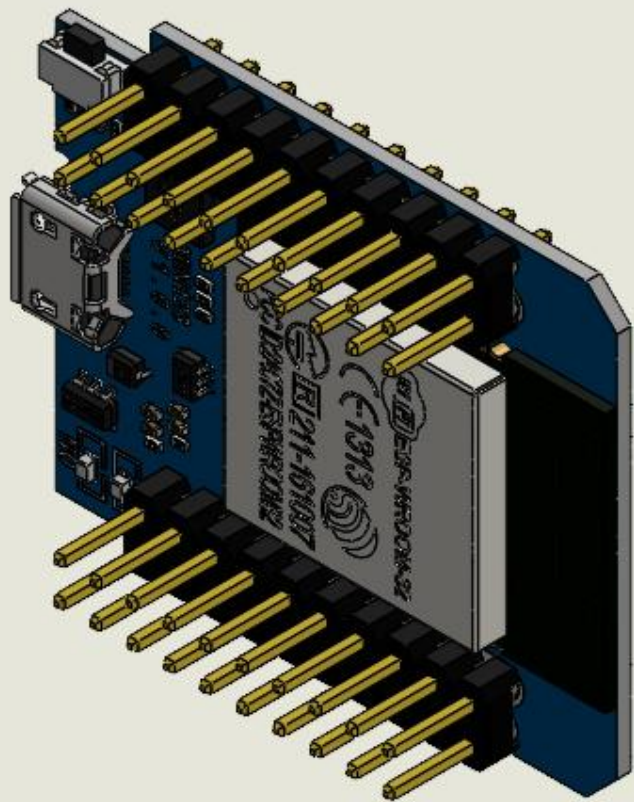
EiD: Me 574

SECTION: Me 574 DATE: 5/4/2025

PART NAME:

Part 19

ESP32-MINI-1



ESP32-MINI-1
*OR EQUIVALENT



DESIGNER: BAILEY WALKE
MASS: lbs
VOLUME: in³
MATERIAL: PURCHASED

DEFAULT TOLERANCE(S)
.X: ±0.1
.XX: ±0.05
.XXX: ±0.001

EiD: ME574

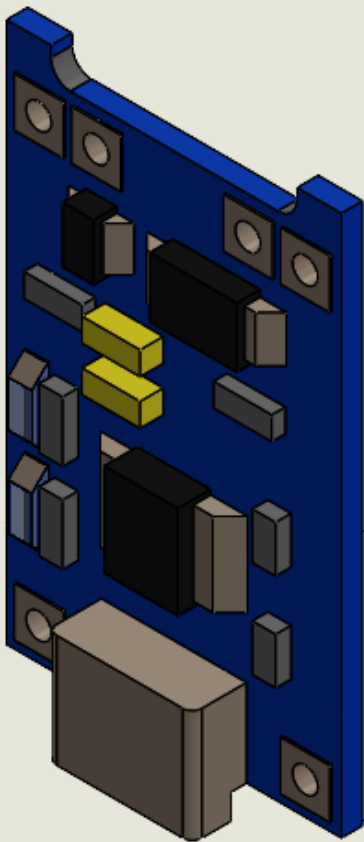
SECTION:

DATE: 5/4/2025

PART NAME:

Part 20

TP4056

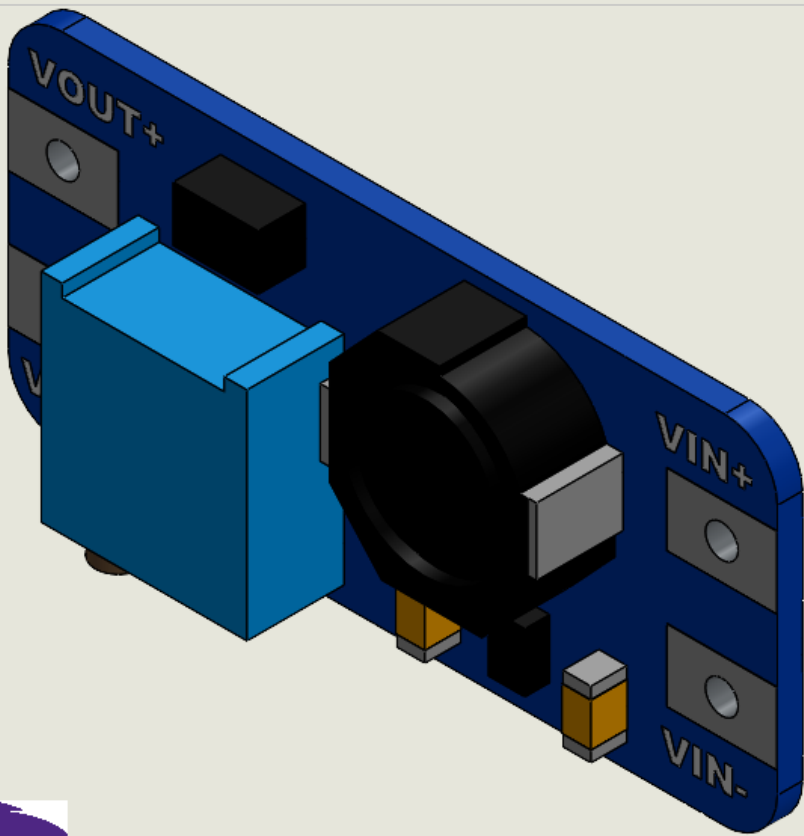


TP4056
*OR EQUIVALENT



DESIGNER: ME 574 Team 3	DEFAULT TOLERANCE (IN)	EiD: Me 574
MASS: lbs	.X: ±0.1	SECTION: Me 574 DATE: 5/4/2025
VOLUME: in³	.XX: ±0.05	PART NAME:
MATERIAL:	.XXX: ±0.001	Part 21

MT3608



MT3608
*OR EQUIVALENT



DESIGNER: ME 574 Team 3

MASS: lbs

VOLUME: in^3

MATERIAL:

DEFAULT TOLERANCE (in)

.X: ±0.1

.XX: ±0.05

.XXX: ±0.001

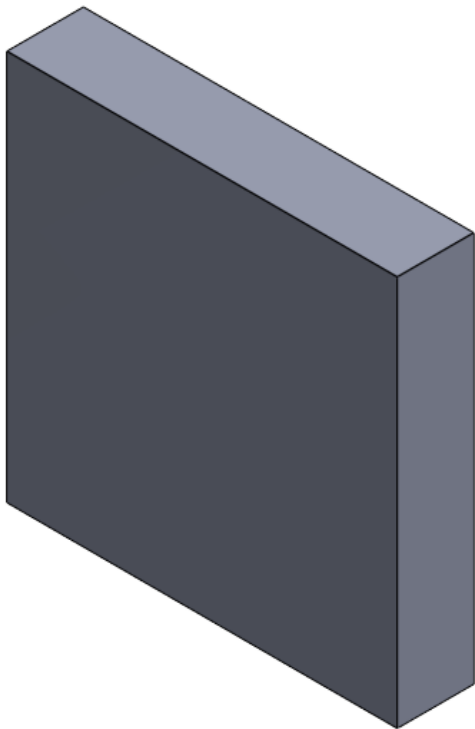
EiD: Me 574

SECTION: Me 574 DATE: 5/4/2025

PART NAME:

Part 22

Battery



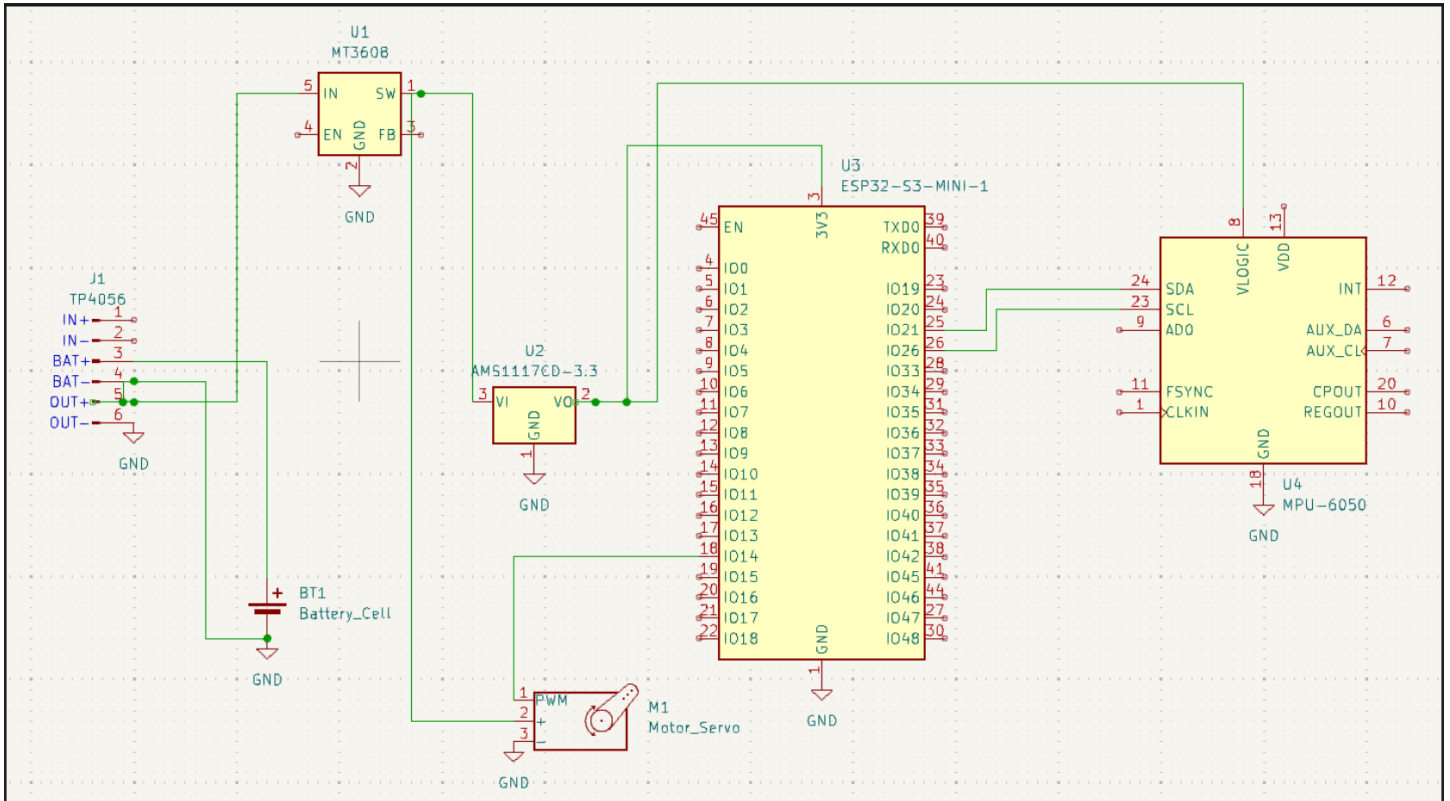
3.7V 2000mAh BATTERY,
2"X2"X0.4" MAXIMUM



DESIGNER: ME 574 Team 3	DEFAULT TOLERANCE (IN)
MASS: lbs	.X: ± 0.1
VOLUME: in^3	.XX: ± 0.05
MATERIAL:	.XXX: ± 0.001

EiD: Me 574	
SECTION: Me 574	DATE: 5/4/2025
PART NAME:	
Part 23	
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Circuit



Code

```
#include <Wire.h>
#include <Adafruit_MPU6050.h>
#include <Adafruit_Sensor.h>
#include <ESP32Servo.h>
#include <math.h>

// === Components ===
Adafruit_MPU6050 mpu;
Servo resistanceServo;

// === Pin Assignments ===
const int servoPin = 25;
const int SDA_pin = 21;
const int SCL_pin = 22;

// === Tremor Detection Settings ===
const int sampleSize = 10;
float tremorData[sampleSize] = {0};
int tremorIndex = 0;

unsigned long lastPeakTime = 0;
unsigned long peakInterval = 0;
float lastGyroMagnitude = 0;
unsigned long lastTremorTime = 0;

const float MIN_TREMOR_THRESHOLD = 0.3; // Ignore very small motion
const float MAX_MACRO_MOVEMENT = 6.0; // Large amplitude = intentional motion
const float MIN_FREQUENCY = 5.0; // Tremors usually above 5 Hz
const float MAX_FREQUENCY = 15.0;
const float MACRO_MOVEMENT_THRESHOLD = 3.5; // Below this = likely intentional
const unsigned long RESET_TIME = 1000; // Time before reset to 0°

int currentAngle = 0;

// === Smooth Servo Move ===
void smoothServoMove(int fromAngle, int toAngle, int stepDelay = 10) {
    if (abs(toAngle - fromAngle) < 3) return;

    if (toAngle > fromAngle) {
        for (int pos = fromAngle; pos <= toAngle; pos++) {
            resistanceServo.write(pos);
            delay(stepDelay);
        }
    } else {
        for (int pos = fromAngle; pos >= toAngle; pos--) {
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        resistanceServo.write(pos);
        delay(stepDelay);
    }
}
currentAngle = toAngle;
}

void setup() {
    Serial.begin(115200);
    delay(100);

    Wire.begin(SDA_pin, SCL_pin);

    if (!mpu.begin(0x68, &Wire)) {
        Serial.println("MPU6050 not detected!");
        while (1);
    }
    Serial.println("MPU6050 Connected.");

    mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
    mpu.setGyroRange(MPU6050_RANGE_500_DEG);
    mpu.setFilterBandwidth(MPU6050_BAND_21_HZ);

    resistanceServo.setPeriodHertz(50);
    resistanceServo.attach(servoPin, 500, 2400);
    resistanceServo.write(0);
}

void loop() {
    sensors_event_t accel, gyro, temp;
    mpu.getEvent(&accel, &gyro, &temp);

    float gyroMagnitude = sqrt(pow(gyro.gyro.x, 2) +
                                pow(gyro.gyro.y, 2) +
                                pow(gyro.gyro.z, 2));

    Serial.print("Gyro magnitude: ");
    Serial.print(gyroMagnitude);

    unsigned long currentTime = millis();

    // Reset after stillness
    if (gyroMagnitude < MIN_TREMOR_THRESHOLD) {
        if (millis() - lastTremorTime > RESET_TIME) {
            smoothServoMove(currentAngle, 0);
            Serial.println(" | No tremor, servo reset to 0°.");
        } else {

```

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    Serial.println(" | Low motion, waiting...");
}
return;
}

// Peak detection
if (abs(gyroMagnitude - lastGyroMagnitude) > 0.1) {
    peakInterval = currentTime - lastPeakTime;
    lastPeakTime = currentTime;
    lastTremorTime = currentTime;
}
lastGyroMagnitude = gyroMagnitude;

// Frequency estimation
float frequency = (peakInterval > 50 && peakInterval < 500) ?
    (1000.0 / peakInterval) : 0;

Serial.print(" | Freq: ");
Serial.print(frequency);
Serial.print(" Hz");

// Ignore intentional movement based on frequency or amplitude
if ((frequency > 0 && frequency < MACRO_MOVEMENT_THRESHOLD) ||
    (gyroMagnitude > MAX_MACRO_MOVEMENT && frequency < 8.0)) {
    Serial.println(" | Voluntary movement ignored.");
    return;
}

// Tremor detected – process and respond
if (frequency >= MIN_FREQUENCY && frequency <= MAX_FREQUENCY) {
    tremorData[tremorIndex] = gyroMagnitude;
    tremorIndex = (tremorIndex + 1) % sampleSize;

    float sum = 0;
    for (int i = 0; i < sampleSize; i++) {
        sum += tremorData[i];
    }
    float avgTremor = sum / sampleSize;

```